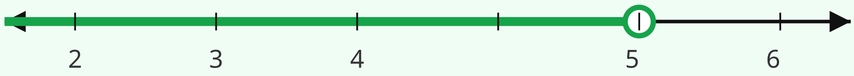
Double Inequalities

AQA · KS4 Year 9–10

Name: ___________________ Class: _________ Date: _________

Solve double (three-part) inequalities such as a < bx + c   d. Keep all three parts balanced, and list the integer solutions or show them on a number line.

VISUAL EXAMPLE



x < 5 on a number line: open circle at 5, shade everything to the left.

COMMON MISTAKES

- Only doing the operation to two parts — you must do it to ALL three parts.
- Forgetting to reverse BOTH inequality signs when dividing all parts by a negative.
- Mixing up which end value is included (watch < versus   ).
- Listing integers wrongly at the ends — check whether each endpoint is included.

METHOD

1. A double inequality has three parts joined by two signs; treat the whole thing together.
2. Undo any + or - by doing it to all three parts.
3. Undo any × or ÷ by doing it to all three parts (reverse both signs if dividing by a negative).
4. Write the solution as a single statement, e.g. -1 < x   3.
5. If asked, list the integer solutions or draw the solution on a number line.

WORKED EXAMPLE

Solve: 1 < 2x + 3   9

Step 1 — Three parts: 1, 2x + 3, 9.

Step 2 — Subtract 3 from all parts: -2 < 2x   6.

Step 3 — Divide all parts by 2 (positive, no flip): -1 < x   3.

Step 4 — Solution: -1 < x   3.

Step 5 — Integers: 0, 1, 2, 3 (-1 is excluded, 3 is included).

1 Solve. (2 marks)

a) $2 < x + 1 < 5$

b) $0 \quad x - 2 \quad 4$

HOW TO START

Do the same to all THREE parts.

For $2 < x + 1 < 5$, subtract 1 from each part: $1 < x < 4$.

ANSWERS

(a) = ____________________

(b) = ____________________

METHOD 1 Three parts — work on all of them | 2 Undo +/- across all three | 3 Undo ×/÷ across all three (flip both if ÷ negative) | 4 Write as one statement $a < x \quad b$ | 5 List integers / draw the line if asked

2 Solve. (2 marks)

a) $4 < 2x < 10$

b) 6   3x   18

HOW TO START

Divide all three parts by the number in front of x.

For $4 < 2x < 10$, divide each part by 2: $2 < x < 5$.

ANSWERS

(a) = ______________________

(b) = ______________________

METHOD 1 Three parts — work on all of them | 2 Undo +/- across all three | 3 Undo ×/÷ across all three (flip both if ÷ negative) | 4 Write as one statement a < x   b | 5 List integers / draw the line if asked

3 Solve. (3 marks)

a) $1 < 2x + 1 \quad 7$

b) $-3 \quad 2x - 1 < 5$

c) $5 \quad 3x + 2 \quad 14$

ANSWERS

(a) = ______

(b) = ______

(c) = ______

METHOD 1 Three parts — work on all of them | 2 Undo +/- across all three | 3 Undo ×/÷ across all three (flip both if ÷ negative) | 4 Write as one statement a < x   b | 5 List integers / draw the line if asked

4 List the integer values of x that satisfy each inequality. (3 marks)

a) -2 < x   3

b) 0   x < 4

c) -3   x   1

ANSWERS

(a) = ______________________

(b) = ______________________

(c) = ______________________

METHOD 1 Three parts — work on all of them | 2 Undo +/- across all three | 3 Undo ×/÷ across all three (flip both if ÷ negative) | 4 Write as one statement a < x   b | 5 List integers / draw the line if asked

5 Solve. (3 marks)

a) 7 < 2x + 1   15

b) -4   3x - 1 < 11

c) 2   (x ÷ 2) + 1 < 5

ANSWERS

(a) = ____________________

(b) = ____________________

(c) = ____________________

METHOD 1 Three parts — work on all of them | 2 Undo +/- across all three | 3 Undo ×/÷ across all three (flip both if ÷ negative) | 4 Write as one statement a < x   b | 5 List integers / draw the line if asked

6 Solve. Take care — you will need to reverse the signs. (3 marks)

a) $1 \quad 5 - x < 4$

b) $-2 < 3 - x \quad 2$

ANSWERS

(a) = ______________________

(b) = ______________________

METHOD 1 Three parts — work on all of them | 2 Undo +/- across all three | 3 Undo ×/÷ across all three (flip both if ÷ negative) | 4 Write as one statement $a < x \quad b$ | 5 List integers / draw the line if asked

7 Solve $4 \quad 2x - 6 < 10$. (4 marks)

a) Write the solution as a single inequality. (2 marks)

b) List the integer values of x. (1 mark)

c) Show the solution on a number line. (1 mark)

ANSWERS

(a) = ______________________

(b) = ______________________

(c) = ______________________

METHOD 1 Three parts — work on all of them | 2 Undo +/- across all three | 3 Undo ×/÷ across all three (flip both if ÷ negative) | 4 Write as one statement $a < x \quad b$ | 5 List integers / draw the line if asked

8 a) A length L is measured as 50 cm to the nearest 10 cm. Write a double inequality (the error interval) for L. (2 marks) (5 marks)

b) Solve $3 \quad (y + 1) \div 2 < 6$ and list the integer values of y. (3 marks)

ANSWERS

(a) = ____________________

(b) = ____________________

METHOD 1 Three parts — work on all of them | 2 Undo +/- across all three | 3 Undo ×/÷ across all three (flip both if ÷ negative) | 4 Write as one statement $a < x \quad b$ | 5 List integers / draw the line if asked

Self-reflection

How confident do you feel with this topic now? (tick one)

Not yet | Getting there | Confident | Really confident

Which question did you find the hardest, and why?

Why must I do every step to all three parts?

One thing I will practise before the next lesson:

Teacher key

Q1: a) 1 < x < 4 b) 2   x   6

Q2: a) 2 < x < 5 b) 2   x   6

Q3: a) 0 < x   3 b) -1   x < 3 c) 1   x   4

Q4: a) -1, 0, 1, 2, 3 b) 0, 1, 2, 3 c) -3, -2, -1, 0, 1

Q5: a) 6 < 2x   14   3 < x   7 b) -3   3x < 12   -1   x < 4 c) 1   x ÷ 2 < 4   2   x < 8

Q6: a) subtract 5: -4   -x < -1; ×(-1) and reverse: 1 < x   4 b) subtract 3: -5 < -x   -1; ×(-1) and reverse: 1   x < 5

Q7: a) add 6: 10   2x < 16   5   x < 8 b) 5, 6, 7 c) filled circle at 5, open circle at 8, shaded between

Q8: a) 45   L < 55 b) ×2: 6   y + 1 < 12   -1: 5   y < 11   integers 5, 6, 7, 8, 9, 10